

Generative & Agentic AI

20 Industry-Flavored Company Micro-Internship Modules

Teach-AI Research & Engineering Team

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Abstract

This document outlines 20 hands-on micro-internship modules designed to bridge the gap between academic GenAI concepts and enterprise production engineering. Each module reframes a core Generative AI, RAG, LangChain, LangGraph, or Model Context Protocol (MCP) pattern into a real-world company engineering assignment complete with industry updates, technical deliverables, evaluation rubrics, and conceptual review tests.

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1 Industry & Internship Index

#	Target Company & Sector	Internship Role Title	Core Technology Scope
1	GlobalTech Solutions (IT Services)	Localization AI Intern	LCEL Chains, Translation Pipelines
2	ShopVerse (Retail & E-Commerce)	Catalog Automation Engineer	Pydantic Structured Outputs
3	ShieldCover Insurance (InsurTech)	Claims Workflow Intern	Parallel & Conditional Chains
4	MedCare Systems (Healthcare)	Clinical RAG Engineer	ChromaDB Vector Search & Embeddings
5	LexisClause Legal (Legal Tech)	Knowledge Systems Intern	Metadata-Filtered RAG Retrieval
6	Vertex National Bank (Banking)	Conversational AI Intern	LangGraph SQLite Checkpoint Memory
7	OmniRoute Logistics (Supply Chain)	Agentic Workflows Intern	ReAct Agent Pattern from Scratch
8	Apex Strategy (Consulting)	AI Advisory Intern	Reflexion Self-Critique Agent Loop
9	Apex Dynamics (Smart Manufacturing)	Multi-Agent Systems Intern	Hierarchical Supervisor Agent
10	QuantCapital (Financial Services)	Governance & Risk AI Intern	Human-in-the-Loop Approval Interruption
11	BioMedix Devices (Medical Equipment)	RAG Evaluation Intern	Text Chunking Comparison Strategy
12	TelcoNet Systems (Telecommunications)	MCP Protocol Engineer Intern	Python MCP Server (File Management)
13	DriveTech Motors (Automotive)	Full-Stack AI Engineer Intern	FastAPI + Gemini + MCP Diagnostics
14	WanderLust Global (Tourism)	Personalization AI Intern	Mem0 Hybrid Long-Term Memory
15	AeroSpace Dynamics (Aerospace)	Flight Ops AI Intern	DeepSeek/Gemini Quantitative Math JSON
16	FlavorCraft Foods (Food & Beverage)	Prompt Engineer Intern	Dual-LLM Comparison & Templating
17	QuickBite QSR (Restaurants)	Cloud AI Intern	Firebase Firestore Chat Persistence
18	SkyWay Travel (Tourism Tech)	TypeScript MCP Developer	Node.js/TypeScript Weather MCP Server
19	PriceWatch Retail (Market Intel)	Web Intelligence Agent Intern	Serper API Web Scraping Agent
20	OmniInsurance (Enterprise Insurance)	Enterprise Agent Architect	Nested Subgraph Architecture

Table 1: Overview of 20 Company Micro-Internships and Technical Focus Areas

2 Detailed Micro-Internship Modules

Module 1: LCEL Chain Building for Automated Localization

1. Title Objective

Localization AI Engineering Intern: Complete work that simulates life on the job. Gain insights into leveraging LangChain as a tool for making informed business decisions.

2. About the Business

Company: GlobalTech Solutions (IT Services & Consulting)

GlobalTech Solutions is a global leader in IT Services & Consulting, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at GlobalTech Solutions. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, LangChain.

The programme will show you what kind of problems are solved at GlobalTech Solutions on a day-to-day basis. Specifically, global IT service leaders need automated localization pipelines to parse technical documentation, generate structured bullet points, and convert them across global offshore languages without manual overhead. This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@globaltech.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, LangChain

4. Tasks & Objectives

Technical Problem Statement: Build a multi-stage LangChain Expression Language (LCEL) chain that takes a complex technical topic (e.g., "Kubernetes Autoscaling"), extracts 3 factual summary points in English, and translates those points into target regional languages (e.g., French or Japanese) in a unified pipeable workflow.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Construct a 3-stage chain (**Topic** → **Fact Generation** → **Translation**).
2. Execute end-to-end runs across 3 distinct technical topics.
3. Submit a working Jupyter Notebook (`chain.ipynb`) and a brief architecture explanation.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Master LCEL composition using the pipe operator (`|`).
- Utilize `ChatPromptTemplate` for dynamic multi-stage prompts.
- Implement `StrOutputParser` for seamless data transformations.

Knowledge Evaluation:

1. What does LCEL stand for and why is it preferred over legacy chain classes?
2. Explain the difference between `RunnableSequence` and explicit `|` syntax.
3. What function does `StrOutputParser` perform in an LCEL pipeline?

Module 2: Structured Product Catalog Ingestion

1. Title Objective

Catalog AI Automation Intern: Complete work that simulates life on the job. Gain insights into leveraging Pydantic as a tool for making informed business decisions.

2. About the Business

Company: ShopVerse Marketplace (Retail & E-Commerce)

ShopVerse Marketplace is a global leader in Retail & E-Commerce, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at ShopVerse Marketplace. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, LangChain, Pydantic.

The programme will show you what kind of problems are solved at ShopVerse Marketplace on a day-to-day basis. Specifically, e-commerce platforms process millions of unstructured product descriptions submitted by third-party sellers. production catalog databases mandate strict, type-safe json formats. This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@shopverse.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, LangChain, Pydantic

4. Tasks & Objectives

Technical Problem Statement: Design a LangChain pipeline using Pydantic models to convert unstructured paragraph descriptions into validated JSON schema entries containing attributes such as `title`, `category`, `price_tier`, and key summary features.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Define a Pydantic schema with appropriate field validations.
2. Process 3 raw seller product descriptions into JSON catalog objects.
3. Submit notebook (`structured_catalog.ipynb`) and generated JSON outputs.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Enforce structured LLM output using Pydantic `BaseModel`.
- Leverage `.with_structured_output()` for *schemabinding*.
- Handle validation errors and missing attributes gracefully.

Knowledge Evaluation:

1. Why is Pydantic structured output superior to asking LLMs to "return valid JSON" in prompt text?
2. How does LangChain handle schema mismatch or parsing exceptions during structured output calls?

Module 3: Parallel Claims Assessment & Conditional Routing

1. Title Objective

Claims Workflow Automation Intern: Complete work that simulates life on the job. Gain insights into leveraging LLMs as a tool for making informed business decisions.

2. About the Business

Company: ShieldCover Insurance (InsurTech)

ShieldCover Insurance is a global leader in InsurTech, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at ShieldCover Insurance. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs.

The programme will show you what kind of problems are solved at ShieldCover Insurance on a day-to-day basis. Specifically, insurtech leaders accelerate claims processing by executing damage evaluation and fraud detection concurrently, followed by automated triage routing. This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@shieldcover.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs

4. Tasks & Objectives

Technical Problem Statement: Build a dual-chain system using `RunnableParallel` for concurrent damage assessment and fraud-risk calculation, combined with `RunnableBranch` to route claims conditionally to either "Auto-Approve" or "Manual Audit".

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Construct a parallel analysis execution node.
2. Implement conditional routing logic based on risk score thresholds.
3. Submit notebook (`parallel_claims.ipynb`) demonstrating both auto-approval and manual-review branches.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Execute concurrent LLM calls with `RunnableParallel`.
- Implement conditional logic routing using `RunnableBranch`.

Knowledge Evaluation:

1. When should developers select parallel execution over sequential chaining?
2. How does `RunnableBranch` evaluate condition predicates to determine route selection?

Module 4: Clinical Protocol RAG Pipeline with ChromaDB

1. Title Objective

Clinical RAG Systems Engineer Intern: Complete work that simulates life on the job. Gain insights into leveraging Vector Databases as a tool for making informed business decisions.

2. About the Business

Company: MedCare Health Systems (Healthcare & Clinical QA)

MedCare Health Systems is a global leader in Healthcare & Clinical QA, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at MedCare Health Systems. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, ChromaDB, Vector Databases.

The programme will show you what kind of problems are solved at MedCare Health Systems on a day-to-day basis. Specifically, hospital networks require context-grounded ai search over proprietary clinical guidelines to eliminate hallucination in medical queries and enable auditability with citations. This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@medcare.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, ChromaDB, Vector Databases

4. Tasks & Objectives

Technical Problem Statement: Develop a Retrieval-Augmented Generation (RAG) system utilizing ChromaDB vector store to answer clinical questions grounded exclusively in internal guideline PDFs, highlighting exact source citations.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Ingest and chunk a sample clinical protocol document.
2. Query ChromaDB for top-k matching passages and return grounded answers with citations.
3. Submit notebook (`clinical_rag.ipynb`) and 3 sample Q&A outputs with citations.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Implement text document chunking and vector embeddings.
- Build vector similarity retrieval workflows using ChromaDB.
- Construct context-injected prompt chains to guarantee factual grounding.

Knowledge Evaluation:

1. Why is RAG preferred over model fine-tuning for dynamic clinical protocol search?
2. What role does cosine similarity distance play during vector store retrieval?

Module 5: Metadata-Filtered Legal Precedent Retrieval

1. Title Objective

Legal Knowledge Systems Intern: Complete work that simulates life on the job. Gain insights into leveraging LLMs as a tool for making informed business decisions.

2. About the Business

Company: LexisClause Legal Partners (Legal Tech)

LexisClause Legal Partners is a global leader in Legal Tech, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at LexisClause Legal Partners. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs.

The programme will show you what kind of problems are solved at LexisClause Legal Partners on a day-to-day basis. Specifically, law firms manage vast legal databases spanning jurisdictions, courts, and filing years. pure semantic search often returns irrelevant precedents from incorrect legal jurisdictions. This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@lexisclause.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs

4. Tasks & Objectives

Technical Problem Statement: Extend a RAG pipeline by tagging document chunks with metadata (`jurisdiction`, `court`, `year`) and performing metadata-filtered vector retrieval to restrict searches strictly to relevant legal domains.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Embed case law documents tagged with court and year metadata.
2. Run comparative queries showing search results with vs. without metadata filtering.
3. Submit notebook (`legal_filtered_rag.ipynb`) showing filtering accuracy.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Embed documents with structured metadata dictionaries.
- Perform exact-match metadata filtering in vector searches.

Knowledge Evaluation:

1. Why does pure semantic search fail when searching multi-jurisdiction legal archives?
2. Describe how vector databases execute metadata filter predicate evaluation.

Module 6: Banking Chatbot with Persistent Memory (LangGraph)

1. Title Objective

Conversational AI Systems Intern: Complete work that simulates life on the job. Gain insights into leveraging LangGraph as a tool for making informed business decisions.

2. About the Business

Company: Vertex National Bank (Banking & FinTech)

Vertex National Bank is a global leader in Banking & FinTech, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at Vertex National Bank. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, LangGraph.

The programme will show you what kind of problems are solved at Vertex National Bank on a day-to-day basis. Specifically, financial customer support bots must recall multi-turn loan details across user sessions. langgraph state graphs with durable sqlite checkpointing ensure state persistence. This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@vertex.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, LangGraph

4. Tasks & Objectives

Technical Problem Statement: Implement a stateful banking support chatbot using LangGraph `StateGraph` and `SqliteSaver` checkpointing that persists loan application memory across script restarts and user disconnects.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Design state schema tracking message history and user account state.
2. Demonstrate thread state persistence by closing and restarting the application session.
3. Submit notebook (`persistent_banking_bot.ipynb`) and memory recall trace.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Understand stateful graph architectures in LangGraph.
- Implement durable session memory using SQLite checkpointers.

Knowledge Evaluation:

1. What is a checkpoint in LangGraph and how does it differ from in-memory arrays?
2. How are thread IDs utilized to isolate separate customer support conversations?

Module 7: ReAct Shipment Tracking Agent from Scratch

1. Title Objective

Agentic Workflow Engineer Intern: Complete work that simulates life on the job. Gain insights into leveraging LangGraph as a tool for making informed business decisions.

2. About the Business

Company: OmniRoute Logistics (Supply Chain & Freight)

OmniRoute Logistics is a global leader in Supply Chain & Freight, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at OmniRoute Logistics. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, LangGraph.

The programme will show you what kind of problems are solved at OmniRoute Logistics on a day-to-day basis. Specifically, freight dispatchers require autonomous agents capable of reasoning step-by-step and calling external live apis (tracking, delay monitoring, eta computation). This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@omniroute.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, LangGraph

4. Tasks & Objectives

Technical Problem Statement: Build a ReAct (Reason + Act) agent loop from scratch using LangGraph that alternates between reasoning steps, tool invocation (mock shipment API), and observation processing until answering user tracking queries.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Define reasoning nodes, tool calling nodes, and conditional execution edges.
2. Test with multi-step freight tracking questions requiring external API queries.
3. Submit notebook (`react_agent.ipynb`) and execution trace log.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Implement the fundamental ReAct cycle (Thought → Action → Observation).
- Connect custom Python functions as tools within an agent graph.

Knowledge Evaluation:

1. Name the 3 phases of the ReAct execution loop.
2. Why is the observation phase critical before the agent takes its next reasoning step?

Module 8: Reflexion Agent Self-Critique Advisory Loop

1. Title Objective

AI Strategy Advisory Intern: Complete work that simulates life on the job. Gain insights into leveraging LangGraph as a tool for making informed business decisions.

2. About the Business

Company: Apex Strategy Consulting (Management Advisory)

Apex Strategy Consulting is a global leader in Management Advisory, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at Apex Strategy Consulting. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, LangGraph.

The programme will show you what kind of problems are solved at Apex Strategy Consulting on a day-to-day basis. Specifically, top-tier management consultancies automate draft deliverable quality control using self-reflection patterns (reflexion), iteratively critiquing and polishing strategy recommendations. This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@apex.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, LangGraph

4. Tasks & Objectives

Technical Problem Statement: Construct a two-node LangGraph loop (Responder & Reviser) with max-iteration control that drafts a market entry brief, evaluates its own output against advisory standards, and refines the draft iteratively.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Build a responder node and critique/reviser node loop.
2. Log each iteration's draft and self-critique score.
3. Submit notebook (`reflexion_consulting.ipynb`) showing at least 2 critique-revision cycles.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Implement self-reflection and iterative critique patterns in LLM graphs.
- Apply bounded iteration control guards to prevent infinite loop execution.

Knowledge Evaluation:

1. What risks exist if a self-critique loop lacks a `MAX_ITERATIONS` limit?
2. How does automated self-critique differ from human-in-the-loop validation?

Module 9: Supervisor Multi-Agent System for Plant Operations

1. Title Objective

Multi-Agent Systems Intern: Complete work that simulates life on the job. Gain insights into leveraging LangGraph as a tool for making informed business decisions.

2. About the Business

Company: Apex Dynamics Industrial (Smart Manufacturing)

Apex Dynamics Industrial is a global leader in Smart Manufacturing, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at Apex Dynamics Industrial. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, LangGraph.

The programme will show you what kind of problems are solved at Apex Dynamics Industrial on a day-to-day basis. Specifically, modern industrial manufacturing plants deploy hierarchical multi-agent teams where a supervisor routes plant breakdown inquiries to specialist agents (searcher, log coder, validator). This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@apex.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, LangGraph

4. Tasks & Objectives

Technical Problem Statement: Construct a hierarchical multi-agent supervisor network in LangGraph where a central Supervisor orchestrates specialized worker agents (Research Agent, Python REPL Log Analyzer, Quality Validator Agent).

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Implement Supervisor, Coder, Researcher, and Validator nodes.
2. Route a mixed query ("Research breakdown causes on Line 3 and analyze sensor logs") end-to-end.
3. Submit notebook (`supervisor_multiagent.ipynb`) and architecture diagram.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Build multi-agent orchestration graphs with dynamic routing.
- Enforce schema-validated task delegation across specialized sub-agents.

Knowledge Evaluation:

1. Why choose a supervisor-worker architecture over a single large monolithic prompt?
2. What quality failures does the Validator node protect against?

Module 10: Human-in-the-Loop Trade Approval Workflow

1. Title Objective

AI Risk & Governance Intern: Complete work that simulates life on the job. Gain insights into leveraging LangGraph as a tool for making informed business decisions.

2. About the Business

Company: QuantCapital Trading (Financial Services)

QuantCapital Trading is a global leader in Financial Services, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at QuantCapital Trading. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, LangGraph.

The programme will show you what kind of problems are solved at QuantCapital Trading on a day-to-day basis. Specifically, autonomous financial trading execution requires mandatory human sign-off checkpoints (hitl) before submitting high-value stock trades or capital transfers. This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@quantcapital.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, LangGraph

4. Tasks & Objectives

Technical Problem Statement: Build a LangGraph workflow incorporating `interrupt()` checkpoints that prepares stock order payloads, halts graph execution for human review, and resumes or halts execution based on manual approval.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Build trade preparation graph with an explicit interrupt state node.
2. Execute two test scenarios: one approved by human input and one rejected.
3. Submit notebook (`hitl_trade_approval.ipynb`) and interactive run logs.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Implement Human-in-the-Loop (HITL) interrupt and resume patterns in LangGraph.
- Ensure regulatory compliance in automated execution systems.

Knowledge Evaluation:

1. Why is state machine interruption preferred over blocking `input()` calls in web applications?
2. How does LangGraph maintain state during interrupted execution sessions?

Module 11: Text Chunking Benchmark for Medical Device Manuals

1. Title Objective

RAG Evaluation & Benchmarking Intern: Complete work that simulates life on the job. Gain insights into leveraging LangChain as a tool for making informed business decisions.

2. About the Business

Company: BioMedix Devices (Medical Equipment)

BioMedix Devices is a global leader in Medical Equipment, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at BioMedix Devices. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, LangChain.

The programme will show you what kind of problems are solved at BioMedix Devices on a day-to-day basis. Specifically, technical support for medical devices demands high precision when searching instructions for use (ifu) manuals. the choice of chunking strategy directly impacts retrieval recall and safety. This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@biomedix.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, LangChain

4. Tasks & Objectives

Technical Problem Statement: Implement and systematically compare 4 text chunking strategies (Fixed-Size, Recursive Character, Sliding Window, and Semantic Chunking) on a complex medical device manual, evaluating downstream QA precision.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Process a sample device manual using 4 distinct chunking splitters.
2. Run 5 benchmark queries across all 4 vector indexes and compare accuracy.
3. Submit notebook (`chunking_benchmark.ipynb`) and comparative analysis report.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Implement advanced text splitting techniques in LangChain.
- Evaluate vector retrieval recall and precision across chunking strategies.

Knowledge Evaluation:

1. Why does semantic chunking often outperform fixed-character splitting for technical manuals?
2. Trade-offs between small chunk sizes (high precision) and large chunk sizes (high context)?

Module 12: Custom MCP Server for Telecom Configuration Management

1. Title Objective

MCP Protocol Developer Intern: Complete work that simulates life on the job. Gain insights into leveraging Model Context Protocol (MCP) as a tool for making informed business decisions.

2. About the Business

Company: TelcoNet Systems (Telecommunications)

TelcoNet Systems is a global leader in Telecommunications, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at TelcoNet Systems. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, Model Context Protocol (MCP).

The programme will show you what kind of problems are solved at TelcoNet Systems on a day-to-day basis. Specifically, telecom network operations centers (nocs) leverage the open model context protocol (mcp) standard to safely expose configuration file operations to ai models via stdio transport. This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@telconet.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, Model Context Protocol (MCP)

4. Tasks & Objectives

Technical Problem Statement: Build a Model Context Protocol (MCP) server in Python using the official MCP SDK exposing tool endpoints (`read_file`, `write_file`, `find_replace`, `list_dir`) scoped to network router configs.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Create an MCP server exposing file tools scoped to a configuration folder.
2. Connect an MCP client (or Claude Desktop) and test tool calls.
3. Submit server code (`mcp_file_server.py`) and client interaction log.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Understand Model Context Protocol architecture (Server, Client, Host).
- Build stdio-based tool servers using decorators and JSON-RPC over stdio.

Knowledge Evaluation:

1. What structural problems does the open MCP standard solve compared to custom tool calling APIs?
2. How does stdio transport work in local MCP server environments?

Module 13: Full-Stack Vehicle Diagnostic App with FastAPI + Gemini + MCP

1. Title Objective

Full-Stack GenAI Intern: Complete work that simulates life on the job. Gain insights into leveraging Gemini / DeepSeek as a tool for making informed business decisions.

2. About the Business

Company: DriveTech Motors (Automotive Services)

DriveTech Motors is a global leader in Automotive Services, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at DriveTech Motors. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, Model Context Protocol (MCP), FastAPI, Gemini / DeepSeek.

The programme will show you what kind of problems are solved at DriveTech Motors on a day-to-day basis. Specifically, modern automotive diagnostic centers connect dealership rest backends with llm reasoning engines and mcp servers to query vehicle telemetry data seamlessly. This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@drivetech.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, Model Context Protocol (MCP), FastAPI, Gemini / DeepSeek

4. Tasks & Objectives

Technical Problem Statement: Build a full-stack REST service in FastAPI where a `/query` endpoint receives vehicle complaints, invokes a Gemini LLM client, connects to an MCP diagnostic server tool, and returns structured diagnosis data.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Implement FastAPI backend connecting MCP vehicle diagnostic tool server.
2. Expose `POST /query` and `GET /tools` endpoints.
3. Submit API backend code and Postman/curl call logs.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Integrate FastAPI, Gemini LLM SDK, and Model Context Protocol (MCP) clients.
- Manage async lifespans and CORS middleware in production backend APIs.

Knowledge Evaluation:

1. Trace request flow: **Frontend** → **FastAPI** → **MCP Client** → **MCP Server** → **LLM**.
2. Why is asynchronous context management essential for MCP client-server lifecycles?

Module 14: Personalization Engine with Mem0 Hybrid Memory

1. Title Objective

Personalization AI Intern: Complete work that simulates life on the job. Gain insights into leveraging Mem0 as a tool for making informed business decisions.

2. About the Business

Company: WanderLust Global (Tourism & Hospitality)

WanderLust Global is a global leader in Tourism & Hospitality, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at WanderLust Global. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, Mem0.

The programme will show you what kind of problems are solved at WanderLust Global on a day-to-day basis. Specifically, next-generation travel concierge bots combine short-term conversational context with long-term vector memory (mem0) to recall persistent user preferences (e.g. seating, dietary needs, hotel tier). This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@wanderlust.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, Mem0

4. Tasks & Objectives

Technical Problem Statement: Develop a hybrid memory travel chatbot using Mem0 and a vector database that automatically extracts, stores, and retrieves traveler preferences across distinct user sessions.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Configure Mem0 memory store with a vector store backend.
2. Demonstrate preference retention across two separated user chat sessions.
3. Submit notebook (`mem0_travel_concierge.ipynb`) and memory retrieval log.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Implement long-term memory extraction and vector indexing using Mem0.
- Combine long-term retrieved memory with active short-term prompt context.

Knowledge Evaluation:

1. How does hybrid memory differ from passing sliding-window chat history?
2. Why is vector search preferred over traditional key-value stores for long-term agent memory?

Module 15: Quantitative Math Reasoning Agent with JSON Output

1. Title Objective

Flight Operations AI Intern: Complete work that simulates life on the job. Gain insights into leveraging Gemini / DeepSeek as a tool for making informed business decisions.

2. About the Business

Company: AeroSpace Dynamics (Aerospace Flight Operations)

AeroSpace Dynamics is a global leader in Aerospace Flight Operations, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at AeroSpace Dynamics. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, Gemini / DeepSeek.

The programme will show you what kind of problems are solved at AeroSpace Dynamics on a day-to-day basis. Specifically, flight operations teams require strict, auditable quantitative calculations (fuel burn rate, payload balance, climb gradient) using reasoning models (deepseek-r1 / gemini 2.0 thinking) with structured json output. This module will attempt to emulate the real-world engineering challenges you will face in production. If you would like to connect with our team to find out more information, please email microinternships@aerospace.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, Gemini / DeepSeek

4. Tasks & Objectives

Technical Problem Statement: Develop a flight operations calculation agent using reasoning-focused models to solve complex multi-step physics/math word problems, enforcing step-by-step JSON outputs for audit logs.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Design prompt requiring structured math reasoning in valid JSON.
2. Evaluate on 5 flight-ops calculation problems and save output JSON files.
3. Submit notebook (`aerospace_math_agent.ipynb`) and generated JSON logs.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Utilize reasoning-distilled models for quantitative problem solving.
- Enforce structured JSON output containing explicit step-by-step reasoning arrays.

Knowledge Evaluation:

1. Why are reasoning-distilled models better suited for multi-step math problems than standard chat models?
2. How should applications handle cases where reasoning models output invalid JSON?

Module 16: Dynamic Prompt Templating & Dual-LLM Benchmarking

1. Title Objective

Prompt Engineering Intern: Complete work that simulates life on the job. Gain insights into leveraging Gemini / DeepSeek as a tool for making informed business decisions.

2. About the Business

Company: FlavorCraft Foods (Food & Beverage R&D)

FlavorCraft Foods is a global leader in Food & Beverage R&D, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at FlavorCraft Foods. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, Gemini / DeepSeek.

The programme will show you what kind of problems are solved at FlavorCraft Foods on a day-to-day basis. Specifically, consumer food r&d teams evaluate competing llm models (e.g. gemini 1.5 vs groq llama-3.3) side-by-side using dynamic prompt templates to compare recipe formulation outputs. This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@flavorcraft.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, Gemini / DeepSeek

4. Tasks & Objectives

Technical Problem Statement: Build a dual-LLM comparison dashboard that formats a single dynamic prompt template and dispatches concurrent queries to two distinct LLM providers, rendering side-by-side outputs.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Construct a shared `ChatPromptTemplate` with dynamic slots.
2. Execute concurrent model calls to Gemini and Groq/Llama endpoints.
3. Submit notebook (`dual_llm_comparison.ipynb`) showing side-by-side transcripts.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Design reusable system and human message prompt templates.
- Benchmark output quality across distinct LLM model backends.

Knowledge Evaluation:

1. Why is decoupling prompt templates from LLM execution logic considered an engineering best practice?
2. What architectural factors cause variance between model responses given identical prompt inputs?

Module 17: Cloud-Native Chat Persistence with Firebase Firestore

1. Title Objective

Cloud Conversational AI Intern: Complete work that simulates life on the job. Gain insights into leveraging Firebase Firestore as a tool for making informed business decisions.

2. About the Business

Company: QuickBite QSR (Quick Service Restaurants)

QuickBite QSR is a global leader in Quick Service Restaurants, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at QuickBite QSR. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, Firebase Firestore.

The programme will show you what kind of problems are solved at QuickBite QSR on a day-to-day basis. Specifically, fast-food chains require cloud-persisted chat sessions across mobile apps and self-service kiosks using scalable nosql databases like google cloud firebase firestore. This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@quickbite.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, Firebase Firestore

4. Tasks & Objectives

Technical Problem Statement: Build an ordering assistant chatbot that saves and retrieves conversation history to/from Firebase Firestore NoSQL collections, allowing users to resume orders across devices seamlessly.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Configure Firebase admin SDK and Firestore collection schemas.
2. Test closing and reopening sessions, verifying chat history reload from Firestore.
3. Submit notebook (`firebase_qsr_chat.ipynb`) and Firestore database console screenshot.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Integrate cloud NoSQL databases (Firestore) for conversation persistence.
- Structure conversational state and history in document-oriented collections.

Knowledge Evaluation:

1. Why choose cloud NoSQL stores (Firestore) over embedded local databases (SQLite) for mobile applications?
2. Describe your proposed Firestore document schema for user chat sessions.

Module 18: TypeScript MCP Weather Server for Node.js

1. Title Objective

TypeScript & Node.js AI Developer Intern: Complete work that simulates life on the job. Gain insights into leveraging TypeScript as a tool for making informed business decisions.

2. About the Business

Company: SkyWay Travel (Tourism Tech)

SkyWay Travel is a global leader in Tourism Tech, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at SkyWay Travel. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, Model Context Protocol (MCP), Node.js, TypeScript.

The programme will show you what kind of problems are solved at SkyWay Travel on a day-to-day basis. Specifically, modern node.js web applications leverage the typescript model context protocol sdk and zod validation schemas to build robust mcp tool servers for client models. This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@skyway.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, Model Context Protocol (MCP), Node.js, TypeScript

4. Tasks & Objectives

Technical Problem Statement: Build a Model Context Protocol server in TypeScript/Node.js using `@modelcontextprotocol/sdk` and `Zod` that exposes a `get_weather_by_city` tool over stdio transport.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Create TypeScript MCP project with tool handlers and Zod validation schemas.
2. Verify stdio tool invocation from an MCP client host.
3. Submit TypeScript server codebase and execution log.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Implement MCP servers in TypeScript/Node.js ecosystems.
- Enforce runtime input validation using Zod schemas for tool parameters.

Knowledge Evaluation:

1. Why is runtime schema validation (Zod) essential for tool inputs in MCP servers?
2. How does stdio transport differ from HTTP SSE transport in MCP implementations?

Module 19: Competitor Price Monitoring Agent with Serper API

1. Title Objective

Market Intelligence AI Intern: Complete work that simulates life on the job. Gain insights into leveraging Serper API as a tool for making informed business decisions.

2. About the Business

Company: PriceWatch Retail (Market Intelligence)

PriceWatch Retail is a global leader in Market Intelligence, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at PriceWatch Retail. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, Serper API.

The programme will show you what kind of problems are solved at PriceWatch Retail on a day-to-day basis. Specifically, e-commerce analytics teams automate price tracking by chaining search apis (serper api), html scraper tools (beautifulsoup), and llm summarization chains. This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@pricewatch.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, Serper API

4. Tasks & Objectives

Technical Problem Statement: Build an automated research agent that takes a pricing query (e.g., "latest discount trends on wireless earbuds"), searches the web via Serper API, scrapes top result pages, and generates a structured competitive report.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Construct research pipeline querying Serper API and scraping top URLs.
2. Synthesize multi-source information into a formatted report with source links.
3. Submit notebook (`market_research_agent.ipynb`) and sample research output report.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Combine Search APIs, web scraping, and LLM text processing into an agent.
- Handle unstructured web content extraction and source citation compilation.

Knowledge Evaluation:

1. Why is a structured Search API preferred over directly scraping search engine result pages?
2. What legal and technical challenges arise when scraping arbitrary web pages?

Module 20: Nested Subgraph Multi-Agent Architecture for Enterprise Claims

1. Title Objective

Enterprise Agent Architect Intern: Complete work that simulates life on the job. Gain insights into leveraging LangGraph as a tool for making informed business decisions.

2. About the Business

Company: OmniInsurance Group (Enterprise Insurance)

OmniInsurance Group is a global leader in Enterprise Insurance, operating across multiple regions with a mission to innovate and deliver long-term stakeholder value. With enterprise-scale operations, the company employs thousands of professionals and is widely recognized as a market leader in its sector.

This job simulation is with the specialized AI and Engineering division at OmniInsurance Group. You will gain insight into how our passionate multi-disciplinary experts solve some of the most complex business problems using cutting-edge solutions, such as Python, Generative AI, LLMs, LangGraph.

The programme will show you what kind of problems are solved at OmniInsurance Group on a day-to-day basis. Specifically, enterprise platforms manage complex workflows by nesting specialized multi-node subgraphs (e.g. damage assessment sub-workflow) inside outer supervisor graphs. This module will attempt to emulate the real-world engineering challenges you will face in production.

If you would like to connect with our team to find out more information, please email microinternships@omniinsurance.com.

3. Tools & Technology Stack

Technologies: Python, Generative AI, LLMs, LangGraph

4. Tasks & Objectives

Technical Problem Statement: Architect a two-level LangGraph system where an outer supervisor graph routes high-level claims processing, invoking a compiled multi-node damage-assessment subgraph as a single modular node.

Timeline: 3-4 hours, Self-paced

Deliverables:

1. Build inner damage-assessment subgraph (Research → Estimate → Summarize).
2. Wrap subgraph and mount as a node inside an outer supervisor graph.
3. Submit notebook (`nested_subgraph_claims.ipynb`) and complete state log.

5. Primary Evaluation Criteria

Key Learning Outcomes:

- Compose complex modular agent architectures using nested LangGraph subgraphs.
- Manage hierarchical state passing and parent-child state synchronization.

Knowledge Evaluation:

1. What architectural advantages does nesting subgraphs offer over expanding a single giant graph?
2. How is state isolated and communicated between outer graphs and inner subgraphs in LangGraph?